

Post-quantum transition for device bounded anonymous credentials: a EUDI wallet application

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Abstract

In 2026, member states of the European Union are expected to have running their own EUDI wallet applications instances compliant with the ARF architecture guidelines. The goal of the ARF architecture is to provide privacy-preserving anonymous credentials to the users, but the current implementation fails. In the meantime, in 2024 the European Union has published a roadmap for the post quantum digital transition of members digital infrastructures.

No current implementation of anonymous credentials is post quantum secure at the moment, with cryptologist and mathematician in top publications claiming that is not possible to build a post quantum anonymous credential scheme with the actual technologies, given hardware constraints and advanced requirements. So we asked ourselves: is it possible to build a post quantum device bounded anonymous credential scheme with the actual technologies, giving hardware constraints and technical difficulties? The answer is **yes**, it is possible.

Moreover, in this thesis we present a post quantum anonymous credentials model that can be deployed with the actual technologies and legacy infrastructures, with NIST standardized post quantum signature schemes. We went further and implemented the full model and run it into android devices in order to benchmark the performance and the results are surprising. We also present some optimizations that could be implemented in order to further improve the performance of the scheme, and we show how to implement them with the actual technologies.

Contents

1	Introduction	2
1.1	Post quantum transition	3
2	EUropean Digital Identity Wallet	3
2.1	Architecture overview	4

2.1.1	Functionalities	4
2.1.2	Roles	4
2.2	Vulnerabilities	4
3	State of the art/Literture overview on device bounded anonymous credentials	4
3.1	Anonymous Credentials	4
3.2	Post quantum anonymous credentials	4
4	Our solution	4
5	Implementation	4
6	Results	4
7	Future work	4
8	Conclusion	4

1 Introduction

Anonymous credentials are a powerful tool for preserving privacy in digital interactions between users and service providers. Anonymous credentials were introduced by David Chaum in 1982 [4], and later formalized by Camenisch and Lysyanskaya in 2001 [1, 2]. They allow users to prove their identity or attributes without revealing any additional information about themselves.

The usual model of anonymous credentials consists of three parties:

- The issuer, a central entity who is responsible for issuing credentials to users and eventually revoking them;
- The user, who asks credentials to the issuer and holds them, and eventually use them to prove their identity or attributes to verifiers.
- The verifier, who is responsible for verifying the user’s credentials without learning any additional information about the user.

A basic scheme of how this entity interacts with each other is represented in figure 1.

In the first approach [7], the issuer was signing user’s credentials using a secure protocol that protects user’s attributes. Credential(s) are then showed to a verifier via zero-knowledge proof of knowledge of a credential(s) whose attributes satisfy some predicate.

Systems based on anonymous credentials have been studied and adopted over the years by big tech companies like Microsoft [8], IBM [3], and are in continuous study and development by the scientific community, facing challenges like performance improvements, revocation lists, legacy infrastructures and hardware constraints, and more recently, post quantum security.

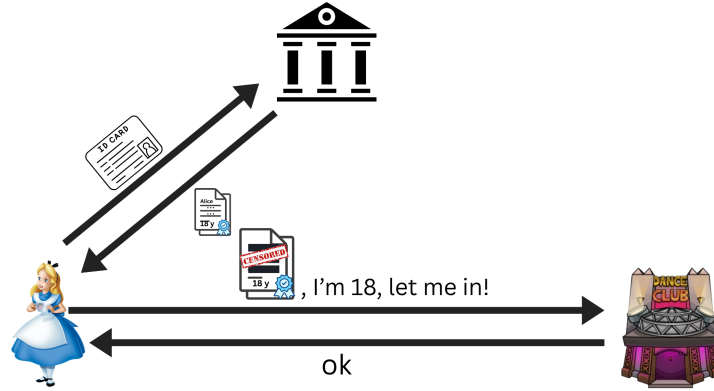


Figure 1: Basic scheme of anonymous credentials.

1.1 Post quantum transition

The european union on 11 April 2024 has published a roadmap for the post quantum digital transition of members digital infrastructures.

The roadmap is a comprehensive plan that outlines the steps that the European Union will take to transition to post-quantum cryptography. The roadmap includes a timeline for the transition, as well as a list of actions that will be taken to ensure a smooth transition.

2 European Digital Identity Wallet

In 2021, the European Commission proposed to adopt a framework for a European Digital Identity [5]. In 2024, the European Commission decided to establish an Architecture Reference Framework (ARF) [6], which consisted of a set of guidelines and properties that each instance of the wallet application should satisfy. This general framework is needed since the commission left freedom to each member state to implement their own instance of the application, due to different possible credentials structures (for example, french and german national identity cards have different structures and/or attributes) and infrastructures (for example, some member states have already a national identity card with an embedded chip, while others don't).

2.1 Architecture overview

2.1.1 Functionalities

2.1.2 Roles

2.2 Vulnerabilities

3 State of the art/Literture overview on device bounded anonymous credentials

3.1 Anonymous Credentials

3.2 Post quantum anonymous credentials

4 Our solution

5 Implementation

6 Results

7 Future work

8 Conclusion

References

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